Southern New Hampshire University

CS-360— Assignment 4-4: Android User Interface Design

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1 Background

A *wireframe* is given to redesign the main screen of the Phyphox (n.d.) mobile phone application— which exploits the many sensors on a smartphone for the measurement of experimental observations in the physical sciences.[1] The purpose of this project is purely pedagogical: it exists for the sake of demonstration of the process of user interface design to a pre-existing, mature mobile phone application

[1](a) These experiments are typically suited for the high school or college undergraduate student; (b) This application targets both the Apple iOS and Android platforms, but I shall focus more on the latter.

in accordance with the guidelines written out in CS-360 (n.d.), whilst allowing for this student to express an alternative user interface design regarding the subject of study.[2]



Figure 1: Comparison between the actual Phyphox (n.d.) main screen (left), and a hypothetical wireframe redesigning the main screen (right).

[2]Just to clarify, Phyphox (n.d.).

2 Wireframe for a Hypothetical Phyphox (n.d.) Main Screen

Figure 1 depicts a comparison of the actual *Phyphox* application's main screen and the wireframe that I made that illustrates the "bare bones" of an alternative design. In particular, figure 1a is the actually published Phyphox application,[3] and figure 1b is the wireframe that I made to describe an alternative design for the Phyphox (n.d.) Android application— which was made with the Pencil Project (n.d.) utility.

The Phyphox (n.d.) mobile phone application has dozens of tools that allow for the measurement of phenomena in the physical environment. There are different categories of measurement, and for each category, there are several tools that somehow exploits the mobile phone's sensors to take measurements based on what category the tool belongs to. To activate a specific tools, one taps on a button with the tool's name and a brief description of its functionality. For example, under the *Acoustics* category, there is a tool called *Audio Amplitude*, which is used to "get the amplitude of sounds." Accessing the "Audio Amplitude" tool is as simple as tapping on its button.

On the main screen of the published Phyphox (n.d.) mobile application for Android phones and tablets,[4] each category is listed on a "master table," with each tool manifested in its respective button, and after each button has been rendered, another category is listed after the last tool's button. The process repeats

[3] I downloaded figure 1a from the Phyphox (n.d.) *Google Play* entry (on Sept. 27, 2026): https://play.google.com/store/apps/details?id=de.rwth_aachen.phyphox&hl=en_US

[4] The Phyphox (n.d.) user interface might be implemented differently on the Apple iOS version of the app, though I am not sure.

itself until the final tool of the final category has been rendered. To reiterate, an illustration of this is shown in figure 1a, and the illustration communicates "what is going on" better than my verbalistic explanations.

The proposed "re-illustration" of the Phyphox (n.d.) app's main screen shows a drop-down menu for each category, a rendering of each button linked to its respective tool, and a text field describing the tool's functionality— with the text field description being independent of the tool's button. The major difference between the actually published Phyphox (n.d.) app and my wireframe is that in order to select a tool from a different category, one must first navigate to the tool's respective category via the drop-down menu, and then select the button corresponding to the tool of interest.

Another difference is the decoupling the tool name from its description: on the actual Phyphox (n.d.) app, the description is embedded into the button, whereas in my wireframe, the description is in its own separate text field. A third notable difference are the themes: the Phyphox (n.d.) app has a "slicker" white-and-orange-on-black foreground/background colour scheme, and the buttons appear to be less "bulky," whereas my wireframe uses simple themes, typography, and colours, but does appear a bit "bulkier" in the amount of space that they take up. To reiterate, figure 1b depicts my wireframe of an alternative user interface for the Phyphox (n.d.) app, and the illustration better "what is going on" than my verbalistic explanations.

2.1 Best Practices for Android UI Design

The user interface is a critical aspect of mobile phone applications with a large target user-base. In particular, the application should be easy to use, make things more convenient for the end-users, actually assist in solving problems in a timely manner, and appeal to the most people in a target population, be they undergraduate students of physical science in the case of Phyphox (n.d.), or "geared" towards more general audiences as in the case of *YouTube* or *Reddit*. I shall use the relevant guidelines in ten guidelines listed in GeeksForGeeks (2025a) as a standard by which to judge the quality of my design choices and justify my decisions.[5]

Regarding good "materials" and "look and feel" of my proposed main screen for the Phyphox (n.d.) app, they are simpler, "bulkier" and more "bland" when compared to the "sleek white-and-orange-on-black" foreground and background themes and colouring. It could be argued that, depending on what subsets of the population that I am targeting for my app, the "sleekness" of Phyphox (n.d.) might be confusing, but further tests will need to be made regarding that claim.

Regarding the ease of navigation, one could argue that my approach of using a drop-down of categories might be slightly more cumbersome and time-consuming compared to the official Phyphox (n.d.) app's approach of just allowing the end-user to "scroll all the way down" until they can identify the category and tool that they want, though this will need to be further tested.

Regarding the accessibility of both the Phyphox (n.d.) and my wireframe of a hypothetical main screen, I did not take accessibility into account in my wireframe, whereas the official authors of Phyphox (n.d.) most likely did, so their

[5]Items 2, 3, 6, 8 and 10 of GeeksForGeeks (2025a) are not relevant to my specific wireframe, so I will skip them for this subsection, though I will touch on them in the subsection on Limitations.

approach may be superior to my approach, though further testing is needed to work out which design choice is more accessible.

Regarding the typography of both the Phyphox (n.d.) app and my wireframe of a hypothetical *Phyphox* main screen, we both use simple fonts, so there is not much difference here. The main difference is not in the penmanship of the rendered text, but rather the colour scheme, where my wireframe is more "traditional," *Microsoft Windows XP*-era typography, and the actual Phyphox (n.d.) app is more "futuristic" in its typography.

Finally, regarding the supporting of visual feedback, I have not included a mechanism for this, so further work will need to be done on this wireframe to improve in this quality.

2.2 Limitations

In the given wireframe, I mention that my designs would need to be "further tested" in many of the points listed in the GeeksForGeeks (2025a) article to gauge the quality of its user-interface designs. A major limitation of my wireframe is that it has not been rigourously tested on a representative sample of its intended user-base. Market research will need to be conducted in order to determine if my approach of rendering tool-buttons dependent on a currently selected "drop-down" menu is, as opposed to the actual Phyphox (n.d.) app's approach of just listing each category after the last tool-button, is, in the subjective judgement of the end-users, a superior approach. These market researches will also need to be done for other aspects of the major differences between the actual Phyphox (n.d.) app, and a "near-complete" prototype following after my wireframe.

Furthermore, the wireframe shown is just a still image. Because of the static nature of a wireframe, I cannot assess specify, nor directly test on a local workstation, the other equally important aspects of mobile application user interface design listed in the GeeksForGeeks (2025a) article: including *responsive design*, *consistency*, *performance optimization*, *collection of user feedback*, and that important good developer habit of ensuring that the app that shall express the wireframe is in accordance with state-of-the-art design. A working prototype will be needed before I can fully judge the quality of a user interface design.

References

CS-360 (n.d.). *Module Four Assignment Guidelines and Rubric*. Southern New Hampshire University.

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